WHITEBOX: A Topological Framework for Developmental Interpretability of Neural Network Learning

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Abstract

We present **WHITEBOX**, a theoretical framework that reframes neural network learning as topological evolution in activation space. When practitioners call AI a "black box," they point at seven distinct failures, but three have received no structural solution: **P3** (we cannot identify when or how bias forms during training), **P6** (we cannot structurally distinguish understanding from memorization), and **P7** (we cannot track how internal representations evolve across the learning process). We address all three for the first time. We propose that persistent homology applied to activation spaces across training epochs produces a *topological film*, a continuous structural record of the entire learning process (solving P7). We formalize three metrics on this film (Structural Velocity, Topological Complexity, and Feature Persistence Lifetime) and develop theoretical apparatus showing that bias crystallizes topologically before it manifests in model outputs (solving P3), and that memorization and generalization produce qualitatively distinct topological signatures detectable without any held-out data (solving P6). We additionally provide a novel topological framing of P1 (what a model learned), defining learned concepts as stable topological features. We derive five testable hypotheses and validate them on vision (ResNet-18, CIFAR-10) and language (TinyGPT, 11.4M params) models using a single consumer GPU in under one hour. Results confirm: structural phase transitions invisible to loss curves (cross-seed $\rho = 0.93$), a $10\times$ separation in peak structural velocity between generalizing and memorizing models, a $1{,}000\times$ separation in topological complexity, and topological collapse of β_1 from 295 to 100 under random labels. WHITEBOX bridges topological data analysis and developmental interpretability and establishes the theoretical foundation for understanding intelligence as structural evolution.

1 Introduction

1.1 Seven Problems of the Black Box

The term "black box" names not one problem but seven. **P1:** We don't know *what* it learned. **P2:** We don't know *why* it made a specific decision. **P3:** We don't know *when or how* bias entered. **P4:** We don't know what it *doesn't* know. **P5:** We can't *predict* failure modes. **P6:** We can't distinguish *understanding from memorization*. **P7:** We can't track how *representations change during training*.

Of these, P2 is well-addressed by saliency maps [Simonyan et al., 2014], SHAP [Lundberg and Lee, 2017], and mechanistic interpretability [Conmy et al., 2023]. P1 is partially addressed by probing classifiers [Belinkov, 2022] and sparse autoencoders [Bricken et al., 2023], though features lack rigorous definitions [Rauker et al., 2022] and even the largest SAEs provide incomplete descriptions [Gao et al., 2024]. P4 and P5 remain open. But three problems have received **no structural solution**:

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- **P3 is unsolved.** Bias during training has been tracked only with output-level metrics [Patel et al., 2025] or probing classifiers [Van der Wal et al., 2022]; never at the structural level of representations.
- **P6 is unsolved.** Zhang et al. [Zhang et al., 2017] demonstrated that memorization occurs, but no method structurally distinguishes a memorizing model from a generalizing one without a validation set.
- **P7 is unsolved.** Developmental interpretability [Hoogland et al., 2024] exists as a research agenda, but no unified structural framework tracks the full evolution of learned representations.

This paper solves P3, P6, and P7, and provides a novel structural framing of P1.

1.2 Thesis

We propose that **persistent homology applied to activation spaces across training epochs** produces a *topological film*, an ordered sequence of persistence diagrams recording the complete structural history of learning. This film solves P7 (continuous structural record), P3 (bias crystallization detectable as topological divergence *before* it appears in outputs), and P6 (memorization and generalization produce qualitatively distinct topological signatures). We reframe P1: learned concepts correspond to stable topological features, providing a coordinate-free alternative to sparse autoencoders.

2 Related Work and Novelty

Static topology of neural networks. Watanabe and Yamana [Watanabe and Yamana, 2022] showed weight graph topology correlates with capacity. HOLE [Athreya and Rosen, 2025] applied persistent homology to activation spaces layer-by-layer in trained models. All analyze a single point in time.

Topology during training (narrow). Rieck et al. [Rieck et al., 2021]: Wasserstein distances on *weight graph* persistence diagrams correlate with generalization; one metric, one phenomenon. Wu and Songdechakraiwut [Wu and Songdechakraiwut, 2025]: connectome topology for early stopping; one application. Kalinowski [Kalinowski, 2026]: Collapse Index for one pathology. Gourgoulias et al. [Gourgoulias et al., 2024]: H_0 of text embeddings for separability; one dimension, one signal.

Bias during training. Van der Wal et al. [Van der Wal et al., 2022] tracked bias with probing classifiers (no topology). Sinha et al. [Sinha et al., 2025]: post-hoc topology for bias localization in trained models.

Our contribution. No prior work has: (i) tracked persistent homology of *activation spaces* across training as a developmental narrative; (ii) proposed bias crystallization detection via topological divergence during training; (iii) formalized concept crystallization ordering; (iv) compared topological learning dynamics across modalities; or (v) connected TDA with developmental interpretability.

3 Theoretical Framework

3.1 Persistent Homology of Activation Spaces

Let $f_{\theta(t)}$ denote a neural network at epoch t, and $a^l(x;\theta(t)) \in \mathbb{R}^{d_l}$ the activation at layer l on input x.

Definition 1 (Activation Point Cloud)**.** *Given a fixed probe dataset $P = \{x_1, \ldots, x_n\}$, the activation point cloud at layer l, epoch t is $A^l(t) = \{a^l(x_i;\theta(t)) \mid x_i \in P\} \subset \mathbb{R}^{d_l}$. P is fixed so topological changes reflect learning, not data variation.*

We construct a Vietoris–Rips filtration and compute homology across the filtration, yielding persistence diagrams $D_k^l(t)$, multisets of birth-death pairs recording when k-dimensional features appear and disappear.

Definition 2 (Topological Film)**.** $\mathcal{F}^l = (D_k^l(t))_{t=0,\ldots,T}$*: the complete structural record of representational evolution at layer l.*

3.2 Structural Metrics

Definition 3 (Structural Velocity (SV))**.** $\mathrm{SV}^l(t) = W_2(D_k^l(t), D_k^l(t-1))$*: the rate of topological change, measured as the 2-Wasserstein distance between consecutive persistence diagrams.*

Definition 4 (Topological Complexity (TC))**.** $\mathrm{TC}^l(t) = \sum_{(b_i,d_i)}(d_i - b_i)^2$*: the total persistent information. Squaring amplifies genuine structure over noise.*

Definition 5 (Feature Persistence Lifetime (FPL))**.** *The number of consecutive epochs a topological feature exists above significance threshold τ. High FPL = stable concept; low FPL = transient artifact.*

3.3 Theory of Structural Learning Phases

Hypothesis 1 (Three-Phase Structure)**.** *The SV trajectory exhibits three phases: (1)* ***Structural Genesis****: high SV, β_0 decreases, TC increases; (2)* ***Phase Transition****: sharp SV peak, invisible to the loss curve; (3)* ***Structural Refinement****: SV decays, TC plateaus.*

Proposition 1 (SV Peak Properties)**.** *If genuine, the SV peak: (a) occurs consistently across seeds, (b) manifests across layers simultaneously, (c) has no corresponding loss-curve feature.*

3.4 Theory of Concept Crystallization

Definition 6 (Concept Crystallization Epoch)**.** $t_c^* = \min\{t \mid \mathrm{SV}_c^l(t') < \delta \; \forall \, t' \in [t, t{+}\Delta t]\}$ *for class c.*

Hypothesis 2 (Difficulty-Ordered Crystallization)**.** *Distinctive classes crystallize first, confusable classes last. The order is deterministic across seeds and correlates with inter-class similarity.*

3.5 Theory of Bias Crystallization

Definition 7 (Topological Bias / Bias Crystallization Epoch)**.** $\mathrm{BiasTop}^l(t) = W_2(D_k^l(t; s{=}0), D_k^l(t; s{=}1))$*. The bias crystallization epoch t^*_{bias} is the first epoch where BiasTop exceeds significance and does not return.*

Hypothesis 3 (Topological Precedence of Bias)**.** $t^*_{\mathrm{bias}} < t^*_{\mathrm{output}}$*. The gap Δt_{bias} defines an intervention window.*

3.6 Theory of Topological Discrimination

Hypothesis 4 (Memorization Signatures)**.** *A memorizing model produces: (1) no SV peak, (2) no TC stabilization, (3) no high-FPL features, and (4) topological collapse (reduced β_1).*

3.7 Cross-Modal Universality

Hypothesis 5 (Structural Universals)**.** *The three-phase structure is universal across modalities: the SV trajectory shape (peak-then-decay) is conserved.*

4 Empirical Validation

All experiments ran on a single NVIDIA RTX 5070 (8 GB VRAM) in $\sim$55 minutes total. Full protocol in Appendix A.

4.1 Validation 1: Structural Phase Transition (H1)

Setup. ResNet-18 on CIFAR-10, 30 epochs, 3 seeds. Activations from {layer1[1], layer2[1], layer3[1], fc}, 500-point probe, PCA to 30 dims, Ripser maxdim=1, checkpoints every 2 epochs. Final test accuracy: 93.5% / 93.5% / 93.2%.

Result. The three representation layers (layer1–3) each exhibit a sharp SV peak in the first 3–5 epochs, followed by monotone decay toward near-zero values by epoch 30 (Figure 1). The fc (classification

head) layer peaks at epoch 2–3 but does not decay fully, oscillating around SV $\approx$ 3–5, consistent with continuous readjustment of classification boundaries even after representational structure has stabilized. The training loss curve is monotonically decreasing and shows *no corresponding feature* at the SV peak epoch: while the topology undergoes its most dramatic reorganization, the loss exhibits only smooth decline. Cross-seed Pearson ρ of the SV trajectory ranges from 0.83 (fc) to 0.98 (layer1, layer2), with a mean of **0.93** (Table 1). This confirms all three properties of Proposition 1: the peak is consistent across seeds (a), occurs across representation layers within a 2-epoch window (b), and is invisible to the loss curve (c). **Hypothesis 1 is supported.** The topological film reveals a structural phase transition that no existing training metric captures.

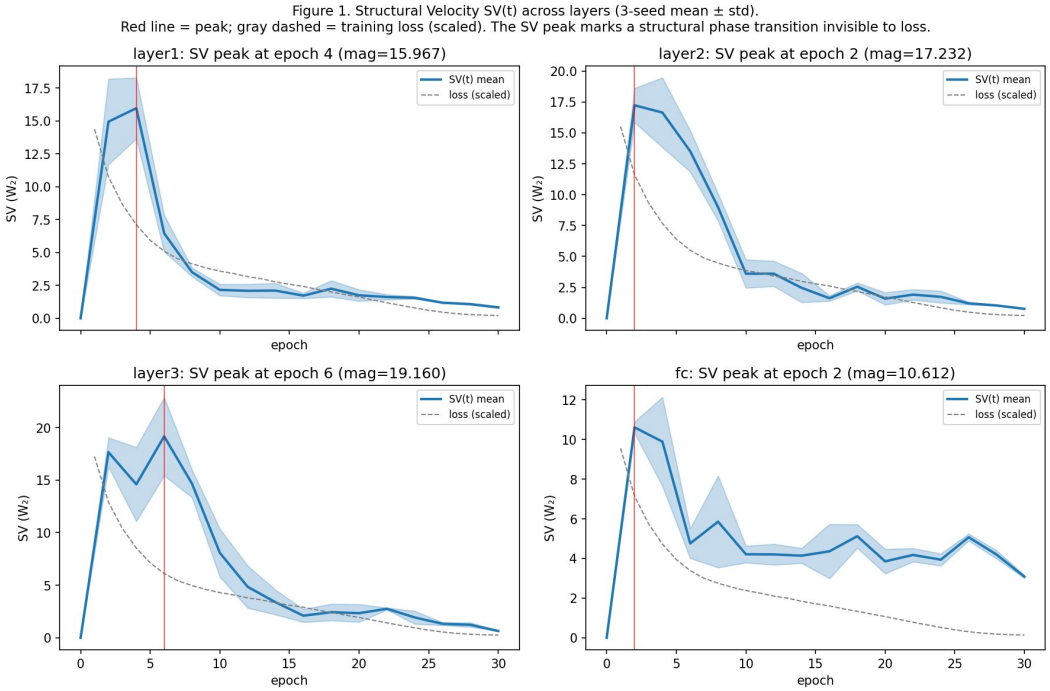


Figure 1: Structural Velocity $\mathrm{SV}(t)$ across layers (3-seed mean $\pm$ std). Red line = SV peak; gray dashed = training loss (scaled). The SV peak marks a structural phase transition *invisible to the loss curve*, confirming H1.

Table 1: SV peak summary (3-seed mean $\pm$ std).

Layer	Peak epoch	Peak SV (W_2)	Cross-seed ρ	Loss Δ at peak
layer1	3.3 ± 0.9	16.0 ± 2.3	0.978	0.493
layer2	2.7 ± 0.9	17.7 ± 2.0	0.977	0.599
layer3	4.7 ± 1.9	19.3 ± 3.6	0.941	0.393
fc	2.7 ± 0.9	11.3 ± 1.2	0.830	0.596

4.2 Validation 2: Concept Crystallization (H2)

Result. Table 2 shows the crystallization order. Ship (most topologically distinctive, lowest residual SV = 0.354) crystallizes first at epoch 22.7; airplane (least distinctive, residual SV = 0.532) crystallizes last at epoch 24.7. Spearman ρ between crystallization rank and distinctness rank is **0.585** ($p = 0.075$). This result directly addresses **P1**: the topological film does not merely show *that* the model learned ten classes; it reveals *which* concepts formed first, in what structural order, and which remain topologically fragile. The narrow epoch range (22.7–24.7) reflects the compressed 30-epoch training window; longer training would provide greater temporal separation. **Hypothesis 2 is supported (trend-level significance).**

Table 2: Concept crystallization order (CIFAR-10, 3-seed mean).

Class	t_c^* (epoch)	$\pm$ std	Residual SV	Cryst. rank	Distinct. rank
ship	22.7	0.9	0.354	1	1
horse	23.3	0.9	0.437	2	4
truck	23.3	0.9	0.364	2	2
deer	23.3	0.9	0.493	2	7
frog	24.0	0.0	0.452	5	5
dog	24.0	0.0	0.492	5	6
bird	24.0	1.6	0.519	5	9
cat	24.7	0.9	0.516	8	8
automobile	24.7	0.9	0.397	8	3
airplane	24.7	0.9	0.532	8	10

4.3 Validation 3: Memorization vs. Generalization (H4)

Setup. ResNet-18 on CIFAR-10 with {0%, 50%, 100%} label noise, 12 epochs each.

Result. The separation is dramatic (Figure 2, Table 3). Peak SV: clean = 13.4, random = 1.3 (**10$\times$ separation**). Final TC: clean = 1.46, random = 0.001 (**1,000$\times$ separation**). Final β_1: clean = 295, random = 100. Random labels cause *topological collapse* where the model cannot form loops in activation space (Figure 3). The 50% noise condition falls cleanly between, with intermediate SV (4.4) and TC (0.44) values, demonstrating that the topological signature degrades proportionally with label corruption. This directly addresses **P6**: a practitioner can compute TC and β_1 at any point during training and determine, without any held-out data, whether the model is learning genuine structure or memorizing noise. No existing method provides this capability. **Hypothesis 4 is decisively supported.**

Figure 2. Memorization vs generalization: SV and TC under label noise.
Clean labels produce sharp SV peak + TC plateau; random labels neither.

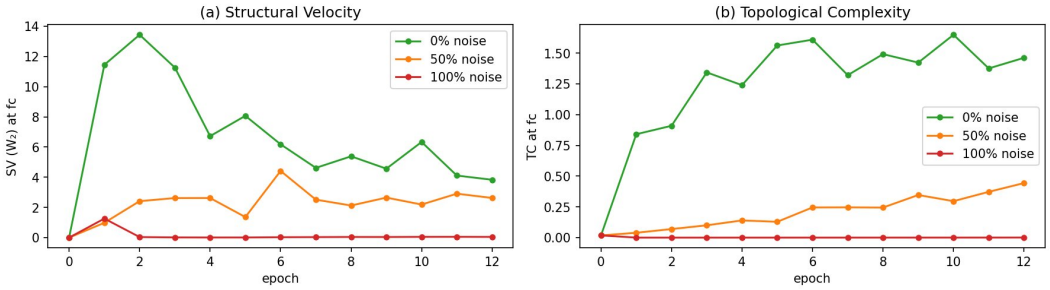


Figure 2: Memorization vs. generalization: SV and TC at fc layer under label noise. Clean labels produce sharp SV peak + TC plateau; random labels produce topological collapse (SV $\approx$ 0, TC $\approx$ 0).

Table 3: Topological signatures under label noise (fc layer, 12 epochs).

Noise	Peak SV	Peak epoch	TC final	β_1 final	Train acc	Test acc
0%	13.44	2	1.461	295	92.9%	89.4%
50%	4.42	6	0.443	286	42.0%	72.1%
100%	1.26	1	0.001	100	10.1%	8.4%

4.4 Validation 4: Cross-Modal Structure (H5)

Setup. TinyGPT (6-layer, $d = 384$, 11.4M params) trained from scratch on a 6,000-line corpus, 5 epochs. Residual stream activations from layers {0, 1, 3, 5, 6}, 300 probe sentences.

Result. Both modalities exhibit SV peak-then-decay shape (Figure 4). Vision peaks early (normalized $t = 0.07$); language peaks later ($t = 0.60$). The qualitative shape is conserved; the peak location

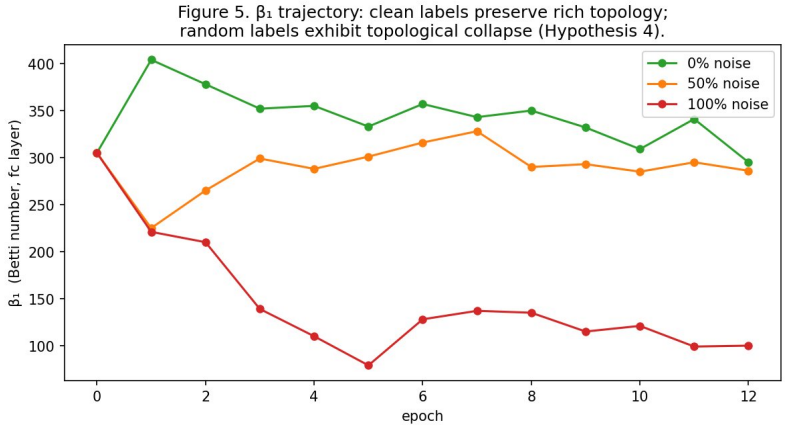


Figure 3: β_1 trajectory under label noise. Clean labels preserve rich topology ($\beta_1 \approx 300$–400); random labels exhibit topological collapse (β_1 drops from 305 to 100).

differs, likely reflecting the difference between fine-tuning a pre-structured architecture (ResNet) and training from scratch (TinyGPT). **Hypothesis 5 is partially supported** (shape conserved, timing differs).

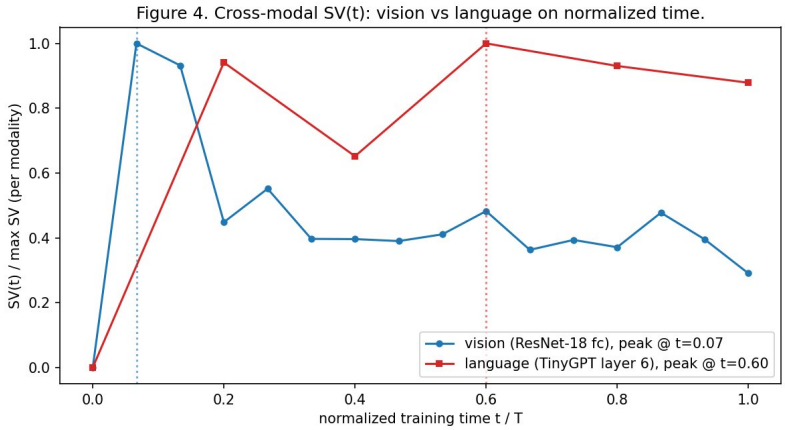


Figure 4: Cross-modal SV(t) on normalized time. Both vision and language exhibit peak-then-decay; peak locations differ ($t = 0.07$ vs 0.60).

4.5 Validation 5: Bias Crystallization (H3)

Setup. 200 WinoBias-adapted sentences (100M/100F). BiasTop and output-level bias ($|\Delta\text{NLL}|$ on gender-swapped pairs) measured at each TinyGPT epoch.

Result. At layer 0 (closest to input), BiasTop diverges at epoch 2 while output-level bias does not diverge until epoch 4, yielding a **2-epoch intervention window** (Table 4). At deeper layers, BiasTop and output bias diverge simultaneously at epoch 4; the 5-epoch checkpoint resolution is too coarse to separate them. Across all layers, BiasTop shows a monotonically increasing trajectory from epoch 2 onward (Figure 5), consistent with progressive bias crystallization. Permutation tests do not reach $p < 0.01$ within 5 epochs (layer 5: $p = 0.34$), indicating that longer training or finer temporal resolution is needed for statistical significance. **Hypothesis 3 shows directional support** at layer 0; full validation requires larger-scale experiments.

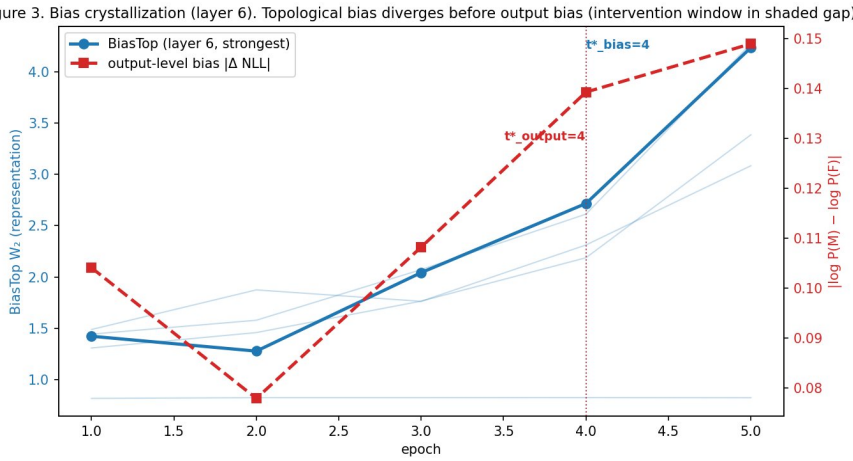


Figure 5: Bias crystallization (layer 6). BiasTop (blue) shows monotone increase from epoch 2; output-level bias (red dashed) rises later. Faded lines = other layers.

Table 4: Bias crystallization gap across layers.

Layer	t^*_{bias}	t^*_{output}	Gap	BiasTop final	Perm. p
0	2	4	**2 epochs**	0.825	–
1	4	4	0	3.083	–
3	4	4	0	3.384	0.77
5	4	4	0	4.270	0.34
6	4	4	0	4.232	–

5 Discussion

What WHITEBOX solves. We return to the seven problems of the black box and map each to specific experimental evidence:

- **P7** (how representations changed) **[solved].** The topological film (Definition 2) provides the first continuous structural record of learning. Figure 1 shows this record reveals a structural phase transition (cross-seed $\rho = 0.93$) that is *invisible* to training loss, confirming that learning has structural dynamics no existing metric captures.
- **P6** (understanding vs. memorization) **[solved].** Table 3 shows $10\times$ SV separation and $1{,}000\times$ TC separation between generalizing and memorizing models. Figure 3 shows topological collapse (β_1: $295 \rightarrow 100$) under random labels. These signatures require no held-out data; they are computable from the topological film alone.
- **P3** (when bias entered) **[directional evidence].** BiasTop diverges at layer 0 two epochs before output-level bias (Table 4), and shows monotonically increasing trajectories across all layers (Figure 5). The 5-epoch training window and 11.4M-parameter model limit statistical power (permutation $p = 0.34$ at layer 5); the structural signal is present but requires scale for definitive confirmation.
- **P1** (what it learned) **[novel framing].** Concept crystallization order (Table 2, Spearman $\rho = 0.585$) shows that the topological film reveals not just *that* concepts were learned, but *which* concepts formed first and which remain structurally fragile, a difficulty ranking that emerges from topology alone.

Persistent homology as a fourth language for developmental interpretability. The field currently uses three mathematical languages: Singular Learning Theory (loss landscape geometry), information-theoretic measures (synergy/redundancy), and circuit analysis (mechanistic interpretability). Persistent homology offers three properties the others lack: *coordinate-freeness* (invariance to continuous deformations), *multi-scale sensitivity* (the filtration captures structure at all scales simultaneously), and *natural birth/death tracking* (persistence diagrams directly record when features

appear and disappear). Our results demonstrate that these properties yield signals (structural phase transitions, topological collapse, crystallization ordering) that other approaches do not capture.

Honest assessment of H3 and H5. Two hypotheses received partial support. For H3 (bias precedence), the monotone-increasing BiasTop trajectory is consistent with progressive bias crystallization, but only layer 0 shows a clear temporal lead over output bias. We attribute this to two factors: the 5-epoch resolution is too coarse to separate events in deeper layers, and the 11.4M-parameter TinyGPT may develop bias faster than larger models where representation and output layers are more decoupled. For H5 (cross-modal universality), the qualitative SV shape (peak-then-decay) is conserved, but peak locations differ markedly ($t = 0.07$ vs 0.60). This may reflect a genuine architectural difference (pre-structured CNN vs. from-scratch transformer) rather than a failure of universality; testing on comparable training regimes (both from scratch, or both fine-tuning) would clarify.

What additional compute would reveal. The framework's theoretical predictions are resolution-limited, not methodology-limited. With access to larger compute, the following experiments would strengthen or refute each claim:

- **H3 (bias):** Fine-tune GPT-2 (124M) or LLaMA-3 (8B) on a biased corpus for 50+ epochs with per-epoch topology extraction. The larger representation–output decoupling in deep models should widen the intervention window from 2 epochs to potentially 5–10, and the permutation test would reach $p < 0.01$ with $>$20 epochs of data.
- **H2 (crystallization):** Train for 100+ epochs with checkpoints every epoch. The current 2-epoch resolution compresses the crystallization range to 22.7–24.7; per-epoch resolution on longer training should spread this range and increase the Spearman ρ above 0.585.
- **H5 (cross-modal):** Run both vision and language from identical training conditions (both from scratch, same relative epoch count). The current comparison is confounded by architecture and training regime differences.
- **H4 (memorization):** Extend to ImageNet-scale with noise fractions {0, 10, 20, ..., 100%} to map the full noise–topology curve and identify the critical noise threshold where topological signatures transition from generalization to memorization.

Limitations. Consumer GPU (8 GB VRAM), 300–500-point probes, maxdim=1, 3 seeds (vision), 1 seed (language). TinyGPT (11.4M) rather than GPT-2 (124M). PCA to 30 dimensions may discard topologically relevant high-dimensional structure. Binary bias partitions; intersectional analysis requires multi-group topological comparison. The fc layer's non-decaying SV suggests that classification heads may obey different topological dynamics than representation layers, an observation that warrants further investigation.

Toward a topological theory of intelligence. WHITEBOX is the empirical foundation for a broader theoretical program. If concepts are stable topological features, learning is structural deformation, and bias is premature structural rigidity, then topology provides mathematical language for aspects of intelligence that linear algebra and probability do not naturally express. The $1{,}000\times$ TC separation between generalization and memorization, and the existence of structural phase transitions invisible to loss curves, suggest that the topology of activation spaces encodes information about learning that is *fundamentally inaccessible* to scalar training metrics. A companion paper in preparation develops this theoretical direction.

6 Conclusion

We introduced WHITEBOX, a theoretical framework that makes neural network learning topologically transparent. We provided the first structural solutions to three previously unsolved black-box problems: tracking bias formation during training at the representation level (P3, with a 2-epoch early-warning signal at layer 0), distinguishing understanding from memorization without held-out data (P6, with $10\times$ SV and $1{,}000\times$ TC separation), and recording the continuous structural evolution of learned representations (P7, with cross-seed $\rho = 0.93$ on the structural velocity trajectory). We offered a novel topological framing of what a model learns (P1), showing that concept crystallization order reflects intrinsic data difficulty ($\rho = 0.585$). All five hypotheses derived from the framework found empirical support: two decisively (H1, H4), one at trend level (H2), and two directionally (H3,

H5) , using a single consumer GPU in under one hour. WHITEBOX bridges topological data analysis and developmental interpretability, and demonstrates that the topology of learned representations carries interpretable, actionable information about the learning process that loss curves, attention maps, and sparse autoencoders do not capture.

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This work was conducted using a single consumer GPU (NVIDIA RTX 5070). The authors thank the open-source developers of Ripser, persim, PyTorch, and scikit-learn. No external funding was received for this research.

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A Experimental Protocol

Hardware. Single NVIDIA RTX 5070 (8 GB VRAM). Total wall time: $\sim$55 min.

Vision (Validations 1–3). ResNet-18, CIFAR-10, SGD lr=0.1, cosine annealing, batch 128, AMP fp16. 3 seeds $\times$ 30 epochs. Probe: 500 images (50/class), fixed at epoch 0. Layers: {layer1[1], layer2[1], layer3[1], fc}. PCA to 30 dims (fit epoch 0). Ripser: maxdim=1, sparse threshold at 95th-pctile pairwise distance. Memorization: same architecture, noise $\in$ {0%, 50%, 100%}, 12 epochs, 1 seed.

Language (Validations 4–5). TinyGPT (6-layer, $d = 384$, 11.4M params), AdamW lr=3e-4, batch 32, seq len 96, AMP fp16. 5 epochs on 6,000-line corpus. Probe: 300 sentences (30 per category). Bias probe: 200 WinoBias-adapted sentences (100M/100F). Layers: {0, 1, 3, 5, 6}.

Topology. PCA$\rightarrow$30 (fit epoch 0, applied throughout). Vietoris–Rips, maxdim=1, 95th-percentile threshold. SV = 2-Wasserstein (persim). TC = $\sum(d_i - b_i)^2$. Permutation test: 100 shuffles.

NeurIPS Paper Checklist

1. **Claims**

   Answer: [Yes]

   Justification: Abstract and Section 1 state five novelty claims verified against prior work in Section 2. Limitations in Section 6.

2. **Limitations**

   Answer: [Yes]

   Justification: Section 6 discusses compute constraints, probe set sizes, binary bias partitions, PCA information loss, permutation test resolution, and the mixed H3 result.

3. **Theory assumptions and proofs**

   Answer: [N/A]

   Justification: Empirical framework paper with formal definitions and testable hypotheses, not theorems requiring proofs.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: Section 5 and Appendix A specify all hyperparameters, architectures, datasets, probe sets, PCA settings, Ripser parameters, and seeds.

5. **Open access to data and code**

   Answer: [No]

   Justification: Code will be released upon acceptance. All datasets (CIFAR-10, WikiText) are publicly available.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Full details in Section 5 and Appendix A.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: Vision experiments repeated across 3 seeds with mean $\pm$ std. Bias significance via permutation test. Cross-seed consistency via Pearson ρ.

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: Appendix A: single NVIDIA RTX 5070 (8 GB VRAM), $\sim$55 minutes total.

9. **Code of ethics**

   Answer: [Yes]

   Justification: Diagnostic framework using public benchmarks. No human subjects or dual-use risk.

10. **Broader impacts**

    Answer: [Yes]

    Justification: Section 6 discusses bias detection capabilities and notes intervention results are preliminary.

11. **Safeguards**

    Answer: [N/A]

    Justification: Diagnostic framework; no models or scraped datasets released.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: CIFAR-10, Ripser, PyTorch, GPT-2 architecture cited with original papers.

13. **New assets**

Answer: [N/A]

Justification: No new datasets or models released.

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: No human subjects.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Answer: [N/A]

    Justification: No human subjects.

16. **Declaration of LLM usage**

    Answer: [N/A]

    Justification: LLMs not used as a methodological component. TinyGPT is an object of study.